

A branch-symmetric Markov family on the RH-366 graph: exact Möbius orthogonality and covariance

RH research program

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Abstract

We derive, rather than geometrically select, a one-parameter Markov/Gibbs family on the primitive four-state graph used by RH-366. In the state order $(--, +-, +-, ++)$, put $q = 1 - t$ and

$$P_t = \begin{pmatrix} t & 0 & q & 0 \\ 1 & 0 & 0 & 0 \\ 0 & t & 0 & q \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad 0 < t < 1.$$

Its stationary law, centered sign variance, and all-lag covariance are exact: the odd covariances vanish and the variance-one covariance at lag $2k$ is $(-q)^k$. Each fixed parameter gives a mixing Markov measure, so the finite-state Chebyshev–Borel–Cantelli argument yields Möbius orthogonality almost surely, simultaneously for all continuous observables. The weighted finite variance has an exact even-shift expansion and a geometric bound. The Parry law is the single member $t = \varphi^{-1}$; the open interval adds non-Parry measures but no canonical arithmetic coupling. All statements are fixed-parameter, and no uniform endpoint theorem, operator trace, zero model, or RH implication is claimed.

1 Frozen graph and derived data type

The source package [1] freezes the state order

$$(--, +-, +-, ++)$$

and the adjacency matrix

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{1}$$

The first coordinate of a state is the scalar sign observable

$$e = (-1, -1, +1, +1)^\top. \tag{2}$$

The graph and its Hénon conjugacy are inherited inputs; they are not reconstructed from the finite calculations below.

For $0 < t < 1$, let $q = 1 - t$ and define the row-stochastic matrix

$$P_t = \begin{pmatrix} t & 0 & q & 0 \\ 1 & 0 & 0 & 0 \\ 0 & t & 0 & q \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{3}$$

Every positive entry of P_t lies on an edge of (1), and every edge of (1) receives a positive probability. Thus (3) is a new derived symbolic family, not a source claim that the Hénon geometry chooses a parameter.

Proposition 1.1 (Stationarity and mixing). *Put $d = 1 + q = 2 - t$. The unique stationary row vector of P_t is*

$$\pi_t = \frac{(1, q, q, q^2)}{d^2}. \quad (4)$$

Moreover,

$$\det(\lambda I - P_t) = (\lambda - 1)(\lambda + q)(\lambda^2 + q). \quad (5)$$

Consequently P_t is primitive and its nontrivial spectral radius is $\sqrt{q} < 1$.

Proof. Multiplying the row in (4) by (3) gives the same row, and its entries sum to one. The directed support is the primitive graph (1); in fact a direct multiplication gives $A^4 > 0$, and all the corresponding entries of P_t^4 are positive for $0 < t < 1$. This proves primitivity and uniqueness of the stationary law. Expanding the determinant in (5) gives the stated factorization. The eigenvalue 1 is simple, while the other three eigenvalues are $-q$ and $\pm i\sqrt{q}$. $\square$

2 Exact observable and covariance

Let ν_t be the stationary Markov measure on Σ_A associated with P_t . With e from (2), direct multiplication gives

$$m_t = \pi_t e = -\frac{t}{d}, \quad v_t = \pi_t (e - m_t \mathbf{1})^2 = \frac{4q}{d^2}. \quad (6)$$

Since $q > 0$, define the centered variance-one observable

$$F_t = \frac{e - m_t \mathbf{1}}{\sqrt{v_t}} = (-\sqrt{q}, -\sqrt{q}, q^{-1/2}, q^{-1/2})^\top. \quad (7)$$

Proposition 2.1 (All-lag covariance). *For every $k \geq 0$,*

$$\text{Cov}_{\nu_t}(F_t, F_t \circ \sigma^{2k+1}) = 0, \quad \text{Cov}_{\nu_t}(F_t, F_t \circ \sigma^{2k}) = (-q)^k. \quad (8)$$

For the raw sign, the even covariance is $v_t(-q)^k$ and the odd covariance is zero.

Proof. The stationary covariance formula for a state observable is

$$\text{Cov}_{\nu_t}(F_t, F_t \circ \sigma^\ell) = \pi_t \text{diag}(F_t) P_t^\ell F_t. \quad (9)$$

Using (3), (4), and (7), one obtains the two exact identities

$$P_t^2 F_t = -q F_t, \quad \pi_t \text{diag}(F_t) P_t F_t = 0. \quad (10)$$

For even $\ell = 2k$, (9)–(10) give $(-q)^k$. For odd $\ell = 2k + 1$, factor out $(-q)^k$ and use the second identity in (10). $\square$

Remark 2.2 (The Parry point). *For $\varphi = (1 + \sqrt{5})/2$, the RH-366 Parry matrix is exactly (3) at*

$$t = \varphi^{-1}, \quad q = \varphi^{-2}. \quad (11)$$

Thus (8) recovers the RH-366 covariance. The theorem is new only in the parameterized non-Parry family and in the fixed-parameter quantifier; it does not rebrand the Parry calculation as a physical spectral result.

3 Fixed-parameter Möbius orthogonality

Theorem 3.1 (Almost-sure orthogonality for each fixed parameter). *For every fixed $t \in (0, 1)$ there is a set $E_t \subset \Sigma_A$ with $\nu_t(E_t) = 1$ such that, for every $\omega \in E_t$ and every $f \in C(\Sigma_A)$,*

$$\frac{1}{N} \sum_{n \leq N} \mu(n) f(\sigma^n \omega) \longrightarrow 0. \quad (12)$$

Under the frozen Hénon conjugacy, the same statement holds for continuous observables on the local survivor.

Proof. It is enough first to treat a centered locally constant function g . The primitive finite-state chain has exponential mixing, so there are constants $C_g < \infty$ and $0 < \rho_t < 1$ such that

$$\left| \text{Cov}_{\nu_t}(g \circ \sigma^i, g \circ \sigma^j) \right| \leq C_g \rho_t^{|i-j|}. \quad (13)$$

With $T_N(\omega) = \sum_{n \leq N} \mu(n) g(\sigma^n \omega)$ and $|\mu(n)| \leq 1$, (13) implies

$$\mathbb{E}_{\nu_t} |T_N|^2 \leq C'_g N. \quad (14)$$

Chebyshev's inequality at $N = k^2$ gives a summable bound for $\nu_t\{|T_{k^2}| > \epsilon k^2\}$. Borel–Cantelli yields $T_{k^2}/k^2 \rightarrow 0$ almost surely. Between consecutive squares there are $O(k)$ bounded summands, so the convergence holds for all N .

Take a countable dense family of centered cylinder functions with rational values and intersect the corresponding full-measure sets. Uniform approximation extends the conclusion to all centered continuous functions. For a general f , subtract its ν_t -mean. The remaining constant term is a multiple of $\sum_{n \leq N} \mu(n) = o(N)$, by the prime number theorem. $\square$

Remark 3.2 (Quantifier firewall). *The set E_t and the constants in (13)–(14) depend on the fixed parameter. No common full-measure set over uncountably many t , and no estimate uniform as $t \downarrow 0$ or $t \uparrow 1$, is asserted.*

4 Möbius-weighted variance

Define

$$S_{N,t}(\omega) = \sum_{n \leq N} \mu(n) F_t(\sigma^n \omega), \quad V_{N,t} = \text{Var}_{\nu_t}(S_{N,t}). \quad (15)$$

The covariance table gives the exact identity

$$V_{N,t} = \sum_{n \leq N} \mu(n)^2 + 2 \sum_{k=1}^{\lfloor (N-1)/2 \rfloor} (-q)^k \sum_{n \leq N-2k} \mu(n) \mu(n+2k). \quad (16)$$

Proposition 4.1 (Unconditional finite bound). *For every $N \geq 1$ and $0 < t < 1$,*

$$0 \leq V_{N,t} \leq \left(1 + 2 \sum_{k \geq 1} q^k \right) N = \frac{2-t}{t} N. \quad (17)$$

Equivalently,

$$\left\| \frac{S_{N,t}}{N} \right\|_{L^2(\nu_t)} \leq \sqrt{\frac{2-t}{tN}}. \quad (18)$$

Proof. Nonnegativity is the variance definition. In (16), bound the diagonal by N , each inner correlation by N , and sum the geometric absolute values. The L^2 estimate is the square root of (17) divided by N . $\square$

Theorem 4.2 (Conditional variance density). *Assume the ordinary fixed-shift two-point correlations*

$$\frac{1}{N} \sum_{n \leq N} \mu(n) \mu(n + 2k) \longrightarrow 0 \quad \text{for every fixed } k \geq 1. \quad (19)$$

Then, for every fixed $t \in (0, 1)$,

$$\frac{V_{N,t}}{N} \longrightarrow \frac{6}{\pi^2}. \quad (20)$$

Proof. The squarefree density gives $N^{-1} \sum_{n \leq N} \mu(n)^2 \rightarrow 6/\pi^2$. For the off-diagonal part of (16), first retain finitely many k and use (19). The remaining tail is bounded uniformly in N by $2 \sum_{k > K} q^k$, which tends to zero as $K \rightarrow \infty$. This proves (20). $\square$

Remark 4.3 (Scope of the conditional statement). *Equation (20) is not an unconditional two-point Chowla theorem. It is a conditional consequence recorded to identify exactly which arithmetic input would be needed for the variance density. The unconditional theorem is (16)–(18).*

5 Parameter boundaries and route decision

At $t = 0$, the branch choices disappear and the chain is a deterministic period-four system; at $t = 1$, the stationary law collapses and $v_t = 0$, so F_t is undefined. Also, the nontrivial spectral radius $\sqrt{1-t}$ tends to one as $t \downarrow 0$, while the entries $q^{-1/2}$ in (7) diverge as $t \uparrow 1$. These are genuine reasons to keep the theorem pointwise on the open interval.

The stationary entropy rate, included only as a structural diagnostic, is

$$h(t) = \frac{-t \log t - (1-t) \log(1-t)}{2-t}. \quad (21)$$

Its maximum occurs at the Parry point (11); this does not select a canonical parameter for the arithmetic problem.

Route	Verdict and boundary
Route A	GO. The derived family has exact stationary, mixing, covariance, almost-sure fixed-parameter Möbius, and finite-variance theorems.
Route B	STOP_SCOPED. The parameter is externally chosen; no canonical arithmetic coupling, determinant, prime-power trace, zeta-zero model, or physical Gate is supplied.

No Hilbert–Pólya operator is constructed, no Riemann zero is identified, and the Riemann hypothesis is not addressed.

6 Executable audit and overlap ledger

The package checks stationarity, the centered P_t^2 eigenidentity, the vanishing one-step covariance, positivity of P_t^4 , and the characteristic covariance table at rational parameters $1/2$, $2/3$, and $3/4$. It also compares (16) with an independent finite double sum through $N = 14$. These are exact finite checks, not asymptotic fits.

The overlap is narrow. RH-366 supplies the graph, sign observable, Parry specialization, and the proof template for almost-sure orthogonality. RH-369 supplies the non-Parry branch-symmetric family and its parameter identities. RH-367 and the PCF parity factor are not used as a continuum operator or as an arithmetic trace. Gates A–E remain false/open.

References

- [1] Source package, *Möbius correlations on a certified Hénon survivor*, commit 34490443f50cfe9af9ff93888e51e7e7e534a5a7, 2026.
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- [3] H. Davenport, On some infinite series involving arithmetical functions (II), *Quart. J. Math.* 8 (1937), 313–320.
- [4] T. Tao, The logarithmically averaged two-point Chowla and Elliott conjectures, *Forum Math. Pi* 3 (2015), e2.